

The Constants of Nature

$$\begin{array}{l}
v_{cs} = \frac{E_1}{2\pi} \left(\frac{t_p}{l_p m_p} \right) \left(1 - \sqrt{\zeta(2)} \, \mathcal{K}_\theta \right) \boxtimes \\
\text{eV : Hz} = \frac{eV}{2\pi} \left(\frac{t_p}{l_p m_p} \right) \left(1 - \sqrt{\zeta(2)} \, \mathcal{K}_\theta \right) \boxtimes \\
\text{J : Hz} = \frac{\text{joule}}{2\pi} \left(\frac{t_p}{l_p m_p} \right) \left(1 - \sqrt{\zeta(2)} \, \mathcal{K}_\theta \right) \boxtimes \\
\text{Hz : eV} = 2\pi \frac{\text{joule}}{\text{eV}} \left(\frac{t_p}{s l_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_\theta \right) \boxtimes \\
h = \frac{\text{joule}}{\text{eV}} \left(\frac{t_p^2}{l_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_\theta \right) \boxtimes \\
h = \left(\frac{t_p^2}{l_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_\theta \right) \boxtimes \\
h = 2\pi \left(\frac{t_p^2}{l_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_\theta \right) \boxtimes \\
q_e = \pi \left(\frac{t_p^2}{l_p m_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_\theta \right) \boxtimes \\
2q_c = 2\pi \left(\frac{t_p^2}{l_p m_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_\theta \right) \boxtimes \\
\frac{\mu_\theta}{e} = \frac{1}{2} \left(\frac{t_p^2}{l_p m_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_\theta \right) \boxtimes \\
\frac{\mu_N}{e} = \frac{1}{2} \left(\frac{t_p^2}{l_p m_p} \right) \left(1 + \sqrt{\zeta(2)} \, \mathcal{K}_\theta \right) \boxtimes \\
G_0^{-1} = \frac{\pi}{\kappa_1^2} \left(\frac{t_p^2}{l_p q_p} \right) \left(1 + \sqrt{\zeta(3)} \, \mathcal{K}_\theta \right) \boxtimes \\
G_0 = \frac{\kappa_1^2}{\pi} \left(\frac{t_p}{l_p m_p} \right) \left(1 - \sqrt{\zeta(3)} \, \mathcal{K}_\theta \right) \boxtimes \\
\text{eV} = \kappa_1 \left(\frac{q_p}{C} \right) \left(1 + \frac{e^{-\gamma}}{6} \, \mathcal{K}_\theta \right) \boxtimes \\
\text{J : eV} = \frac{\text{joule}}{\kappa_1} \left(\frac{C}{q_p} \right) \left(1 - \frac{e^{-\gamma}}{6} \, \mathcal{K}_\theta \right) \boxtimes \\
-\frac{e}{m_e} = -\kappa_1 \left(\frac{q_p}{m_1} \right) \left(1 + \frac{e^{-\gamma}}{6} \, \mathcal{K}_\theta \right) \boxtimes \\
\frac{e}{m_+} = \kappa_1 \left(\frac{m_+}{m_1} \right) \left(1 + \frac{e^{-\gamma}}{6} \, \mathcal{K}_\theta \right) \boxtimes \\
\frac{\mu_\theta}{h} = \frac{1}{4\pi} \text{tesla}^2 \left(\frac{q_p}{m_e} \right) \left(1 + \frac{e^{-\gamma}}{6} \, \mathcal{K}_\theta \right) \boxtimes \\
\frac{\mu_N}{h} = \frac{\kappa_1}{4\pi} \frac{1}{\text{MHz}} \left(\frac{q_p}{s m_1} \right) \left(1 + \frac{e^{-\gamma}}{6} \, \mathcal{K}_\theta \right) \boxtimes \\
a = \frac{4\pi}{32 s \kappa_1^4} \left(\frac{1}{m_e} \right) \left(1 + 35 \left(\frac{4\pi}{8} \right)^5 \mathcal{K}_\theta \right) \boxtimes \\
d_{220} = \frac{1}{\sqrt{8}} \frac{4\pi}{32 s \kappa_1^4} \left(\frac{1}{m_e} \right) \left(1 + 35 \left(\frac{4\pi}{8} \right)^5 \mathcal{K}_\theta \right) \boxtimes \\
R_m = \frac{\kappa_1^4}{4\pi} \left(\frac{m_e}{l_p m_p} \right) \left(1 - \frac{6}{V_0} \right)^2 \mathcal{K}_\theta^3 \boxtimes \\
E_h : \frac{1}{\text{m}} = \frac{2\pi}{\kappa_1^4} \left(\frac{m_+}{l_p m_p} \right) \left(1 - \frac{6}{V_{fe}} \right)^2 \mathcal{K}_\theta^2 \boxtimes \\
\frac{1}{\text{m}} : E_h = \frac{2\pi}{\kappa_1^4} \left(\frac{l_p m_p}{m m_e} \right) \left(1 + \frac{6}{V_{fe}} \right)^2 \mathcal{K}_\theta^2 \boxtimes \\
A_{90} = \text{ampere} \left(1 + \frac{1}{\delta} \left(\frac{4\pi}{35} \right) \mathcal{K}_\theta^3 \right) \boxtimes \\
C_{90} = \text{coulomb} \left(1 + \frac{1}{\delta} \left(\frac{4\pi}{35} \right) \mathcal{K}_\theta^3 \right) \boxtimes \\
\mu_B = \frac{\kappa_1}{4\pi} \left(\frac{t_p q_p}{l_p m_p} \right) \left(1 + \delta \left(\frac{K}{6} \right) \mathcal{K}_\theta^3 \right) \boxtimes \\
\mu_N = \frac{\kappa_1}{4\pi} \left(\frac{t_p q_p}{l_p m_p} \right) \left(1 + \delta \left(\frac{K}{6} \right) \mathcal{K}_\theta^3 \right) \boxtimes \\
\kappa = \left(\frac{t_p^2}{l_p^2} \frac{m_p}{q_p^2} \right) \left(1 + \left(\frac{C_d}{6} \right) \mathcal{K}_\theta^4 \right) \boxtimes \\
A_{perm} = \left(\frac{t_p^2}{l_p^2} \frac{q_p^2}{m_p} \right) \left(1 - \left(\frac{C_d}{6} \right) \mathcal{K}_\theta^4 \right) \boxtimes \\
\epsilon_0 = \frac{1}{4\pi} \left(\frac{t_p^2}{l_p^3} \frac{q_p^2}{m_p} \right) \left(1 - \left(\frac{C_d}{6} \right) \mathcal{K}_\theta^4 \right) \boxtimes \\
A_{time} = \left(\frac{1}{\kappa_1^4} \frac{t_p m_p}{m_e} \right) \left(1 + \sqrt{5} \left(\frac{C_d}{C_0} \right) \mathcal{K}_\theta^4 \right) \boxtimes \\
\text{Hz : } E_h = \text{watt} \frac{2\pi}{\kappa_1^4} \left(\frac{t_p m_p}{m_e} \right) \left(1 + \sqrt{5} \left(\frac{C_d}{C_0} \right) \mathcal{K}_\theta^4 \right) \boxtimes \\
E_h : \text{Hz} = \frac{\kappa_1^4}{2\pi} \left(\frac{m_e}{l_p m_p} \right) \left(1 - \sqrt{5} \left(\frac{C_d}{C_0} \right) \mathcal{K}_\theta^4 \right) \boxtimes \\
R_{\omega C} = \frac{\kappa_1^4}{4\pi} \left(\frac{m_e}{t_p m_p} \right) \left(1 - \sqrt{5} \left(\frac{C_d}{C_0} \right) \mathcal{K}_\theta^4 \right) \boxtimes \\
A_{modm} = \kappa_1 \left(\frac{l_p^2}{l_p} \frac{q_p m_p}{m_p} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \mathcal{K}_r^2 \right) \boxtimes \\
\mu_B = \frac{\kappa_1}{2} \left(\frac{l_p^2}{l_p} \frac{q_p m_p}{m_p} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \mathcal{K}_r^2 \right) \boxtimes \\
\mu_N = \frac{\kappa_1}{2} \left(\frac{l_p^2}{l_p} \frac{q_p m_p}{m_p} \right) \left(1 + \omega_2 \left(\frac{2\pi}{35} \right) \mathcal{K}_r^2 \right) \boxtimes \\
\mu_{tri} = \kappa_1^2 \mathcal{K}_2^2 \left(D_{D0} + \pi \right) \left(\frac{l_p^2}{l_p} \frac{q_p m_p}{m_p} \right) \left(1 - \omega_2 \left(\frac{2\pi}{18} \right) \mathcal{K}_r^2 \right) \boxtimes \\
\frac{\mu_{tri}}{\mu_N} = S \left(D_{D0} + \pi \right) \left(\frac{m_+}{m_{tri}} \right) \left(1 + \frac{4}{L_{Li}} \left(\frac{32}{45} \right) \mathcal{K}_r^2 \right) \boxtimes \\
\frac{\mu_{tri}}{\mu_N} = \kappa_1 \mathcal{K}_2^2 \left(2 D_{D0} + 2\pi \right) \left(\frac{m_+}{m_{tri}} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathcal{K}_r^2 \right) \boxtimes \\
\frac{\mu_{tri}}{\mu_B} = \kappa_1 \mathcal{K}_2^2 \left(2 D_{D0} + 2\pi \right) \left(\frac{m_+}{m_{tri}} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathcal{K}_r^2 \right) \boxtimes \\
g_{tri} = \kappa_1 \mathcal{K}_2^2 \left(4 D_{D0} + 4\pi \right) \left(\frac{m_+}{m_{tri}} \right) \left(1 - \frac{18}{8} \left(\frac{4\pi}{35} \right) \mathcal{K}_r^2 \right) \boxtimes \\
\mu_+ = -\frac{S}{C_{CFP}} \left(\frac{m_+}{m_n} \right) \left(1 + 15 \left(\frac{2\pi}{8K} \right) \mathcal{K}_r^2 \right) \boxtimes \\
\frac{\mu_+}{\mu_n} = -\frac{C_{CFP}}{S} \left(\frac{m_n}{m_+} \right) \left(1 + 15 \left(\frac{2\pi}{8K} \right) \mathcal{K}_r^2 \right)^{-1} \boxtimes \\
\frac{\mu_+}{\mu_n} = \frac{C_{CFP}}{P_{up}} \left(\frac{m_n}{m_+} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_r^2 \right) \boxtimes \\
\frac{\mu_+}{\mu_e} = \frac{P_{up}}{C_{CFP}} \left(\frac{m_+}{m_n} \right) \left(1 + 18 \left(\frac{4K}{2\pi} \right) \mathcal{K}_r^2 \right)^{-1} \boxtimes \\
\mu_e = \left(\frac{m_e}{m_e} \right) \left(1 - \left(\frac{4\pi}{18} \right) \mathcal{K}_r^3 \right) \boxtimes \\
G = \left(\frac{l_p^3}{t_p^2} \right) \left(1 - \left(\frac{4\pi}{18} \right) \mathcal{K}_r^3 \right) \boxtimes \\
\bar{G} = \text{GeV}^2 \left(\frac{t_p^4}{l_p^4 m_p^2} \right) \left(1 - \left(\frac{4\pi}{18} \right) \mathcal{K}_r^3 \right) \boxtimes \\
m_n = \left(\frac{2\pi^2}{25} \right) \frac{\text{GeV}}{\varrho^{-2}} \left(\frac{t_p^2}{l_p^2} \right) \left(1 - \left(\frac{2s}{4\pi} \right) \mathcal{K}_r^3 \right) \boxtimes \\
m_+ = \left(\frac{2\pi^2}{s} \right) \frac{\text{GeV}}{\omega_2} \left(\frac{t_p^2}{l_p^2} \right) \left(1 - \frac{\sqrt{2}}{e^{7/3}} \left(\frac{18}{25} \right) \mathcal{K}_r^3 \right) \boxtimes \\
m_\mu = \frac{2\pi}{35} \left(\frac{4\pi}{6s} \right) \left(A_{mass} \right) \left(1 - 18 K_{-6} \mathcal{K}_r^3 \right) \boxtimes \\
\mu_{de} = -P_{up} \left(\frac$$

$t_p = 5.39125836832313 \times 10^{-38}$ s	Planck time
$l_p = 1.61625918175645 \times 10^{-38}$ m	Planck length
$q_p = 1.87554596713962 \times 10^{-38}$ C	Planck charge
$T_p = 1.41678698590795 \times 10^{32}$ K	Planck temperature
$m_p = 2.1766268317579 \times 10^{-27}$ kg	Planck mass
$\boxtimes = 0.99999199973626 \times 10^{-10}$ ⁷	hyperbolic inversion boundary
$\boxtimes = 0.99999199973626 \times 10^{-10}$ ⁵	number of dimensions
$\boxtimes = 0.99999199973626 \times 10^{-10}$ ⁷	break in scale symmetry
$\kappa_1 = 0.08542454315533004 \dots$	1 st hyperbolic partition constant
$\kappa_2 = 3.66756753485501 \dots$	2 nd hyperbolic partition constant
$\kappa_3 = 1.8764693000417 \dots + 0.06615262615972 \dots i$	3 rd hpc
$\kappa_4 = 1.8764693000417 \dots - 0.06615262615972 \dots i$	4 th hpc
$\kappa_5 = 4.47826244916751 \dots$	hyperbolic radius constant
$\kappa_6 = 2.00316562310924 \dots$	hyperbolic radian constant
$G_{01} = 1.01494160640965 \dots$	Gieseking's constant
$V_{02} = 0.29198321281930 \dots$	figure eight knot hyperbolic volume
$K = 0.0258659417271910 \dots$	Catalan's constant
$C_d = 0.291560904030813 \dots$	domino tiling constant
$D_d = 1.72616281206959 \dots$	dimer constant
$\omega_1 = 0.7649770185288596 \dots + 1.32497062714087 \dots i$	omega_1 constant
$\omega_2 = 1.529954037057192 \dots$	omega_2 constant
$t^{(i)} = 0.438282936727032 \dots + 0.360592471871385 \dots i$	power tower of i
$\varphi_i = (-1)^{1/2}$	imaginary unity
$\varphi = 1.61803398874989 \dots$	golden ratio
$\pi = 3.14159265358979 \dots$	Archimedes' constant
$P_{00} = 2.29558713992963 \dots$	universal parabolic constant
$S = 5.24411510858423 \dots$	arc length of the unit lemniscate
$L = 2.62205755429219 \dots$	lemniscate constant
$L_1 = 1.3110287714605 \dots$	1 st lemniscate constant
$L_2 = 0.599070117367796 \dots$	2 nd lemniscate constant
$U = 0.847213084793979 \dots$	ubiquitous constant
$G_{06} = 0.884626841674073 \dots$	Gauss's constant
$W_{We} = 0.474949398798920 \dots$	Weierstrass constant
$C_{CF} = 0.697774657964819 \dots$	continued fraction constant
$C_{01} = 0.998136044598509 \dots$	Ramanujan's 1 st continued fraction constant
$L_{Luj} = 1.011001000000000 \dots$	Liouville's constant
$M = 1.247156459463318 \dots$	Madelung constant
$P = 1.22674201072035 \dots$	prime constant
$e = 2.71828182845904 \dots$	Euler's number
$\gamma = 0.577215664901532 \dots$	Euler-Mascheroni constant
$S = 0.828252949678807 \dots$	Sierpinski's constant
$D_{DD} = 0.729085133215160 \dots$	Dottie number
$L_1 = 0.662743419439181 \dots$	Laplace limit
$C_{LCP} = 1.19967864025773 \dots$	real fixed point of the hyperbolic cotangent
$J_{01} = 2.404825557695772 \dots$	1 st root of the Bessel function
$\alpha_F = 2.60290787509589 \dots$	alpha Feigenbaum constant
$\alpha_F = 4.56920160910299 \dots$	delta Feigenbaum constant
$S = 3.24697960371714 \dots$	silver constant
$P = 1.3247179572447 \dots$	plastic constant
$C_Q = 0.605443657196732 \dots$	QRS constant
$\kappa_{00} = 2.29316628741186 \dots$	1 st Foias constant
$K_0 = 1.156552374442151 \dots$	Kinchin mean of order -6
$T = 2.73150000000000 \times 10^2$ kelvin	freezing point of water
$p_0 = 1.00000000000000 \times 10^5$ pascal	1 bar
$p_1 = 1.01325000000000 \times 10^5$ pascal	standard atmospheric pressure
$E_1 = 5.90110229711386 \times 10^{-24}$ joules	energy associated with D_{cs}
$f_1 = 6.40000000000000 \times 10^{14}$ Hz	frequency of max light conversion
$\lambda_{peak} = 4.96511423174427 \dots$	peak 2 of spectral radiance/unit wavelength
$\nu_{peak} = 2.821493971212207 \dots$	peak emission of spectral flux/unit frequency
nl = derangement function	
ln = factorial function	$lm = \text{lumen} = \text{cd} \cdot \text{sr}$
$\log(x)$ = natural logarithm	$N = \text{newton} = \text{m kg/s}^2$
$\Gamma(x)$ = gamma function	$\mathcal{N} = \text{noether} = \text{m kg/s}$
$\zeta(x)$ = Riemann zeta function	$\Omega = \text{ohm} = \text{m}^2 \text{ kg/s}^2$
units	$\text{Pa} = \text{pascal} = \text{kg/m}^2 \text{ s}^2$
s = second	$S = \text{siemens} = \text{S} = \text{C}^2/\text{m}^2 \text{ kg}$
m = meter	$T = \text{tesla} = \text{kg/C} \cdot \text{s}$
C = coulomb	$V = \text{volt} = \text{m}^2 \text{ kg/s}^2 \text{ C}$
K = kelvin	$W = \text{watt} = \text{m}^2 \text{ kg/s}^3$
$\text{kg} = \text{kilogram}$	$\text{Wb} = \text{weber} = \text{m}^2 \text{ kg/s} \cdot \text{C}$
derived units	
$\text{MHz} = \text{megahertz} = 1.00000000000000 \dots \times 10^6$ 1/s	
$\text{fm} = \text{femtometer} = 1.00000000000000 \dots \times 10^{-15}$ m	
$\text{eV} = \text{electron-volt} = 1.00000000000000 \dots$ eV	
$\text{MeV} = \text{megaelectron-volt} = 1.00000000000000 \dots \times 10^6$ eV	
$\text{GeV} = \text{gigaelectron-volt} = 1.00000000000000 \dots \times 10^9$ eV	
$\frac{r_a}{r_b} = \left(\frac{4\pi}{15}\right)^2$	$\frac{r_a}{r_b} = K$
$\frac{r_a}{r_b} = \sqrt{\frac{C_1}{C_2}}$	$\frac{r_a}{r_{de}} = \sqrt{\log(2\pi)}$
$\frac{r_a}{r_b} = \sqrt{\frac{C_1$	